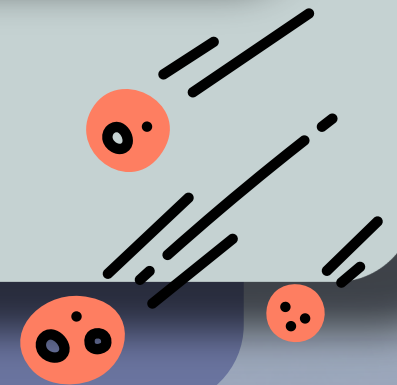


03 Interceptors

Swarm Deployment & Coordination

Presented by : Joseph, Tze Kai, Junyu, Alicia

Mentor: Mr. Chiko David



Current Problem

How can we neutralize 100s of hostile drones during a total communication blackout (jamming)?

How can our **interceptor** drones **make intelligent decisions without communications**?

Coordination without communications (GNSS-denied)

What exists today?

- RF Jammer
- GNSS Spoofer
- Laser
- Pilot-flown Interceptor

What AirDnD adds:

Each lightweight kamikaze drone with small payload observes motion, detects allies & enemies

ZERO GROUND LINKS, ZERO MESH

Demo!

Measured against independent greedy pursuit

100

hostiles per run

125

interceptors

30

paired seeds

Intended Users

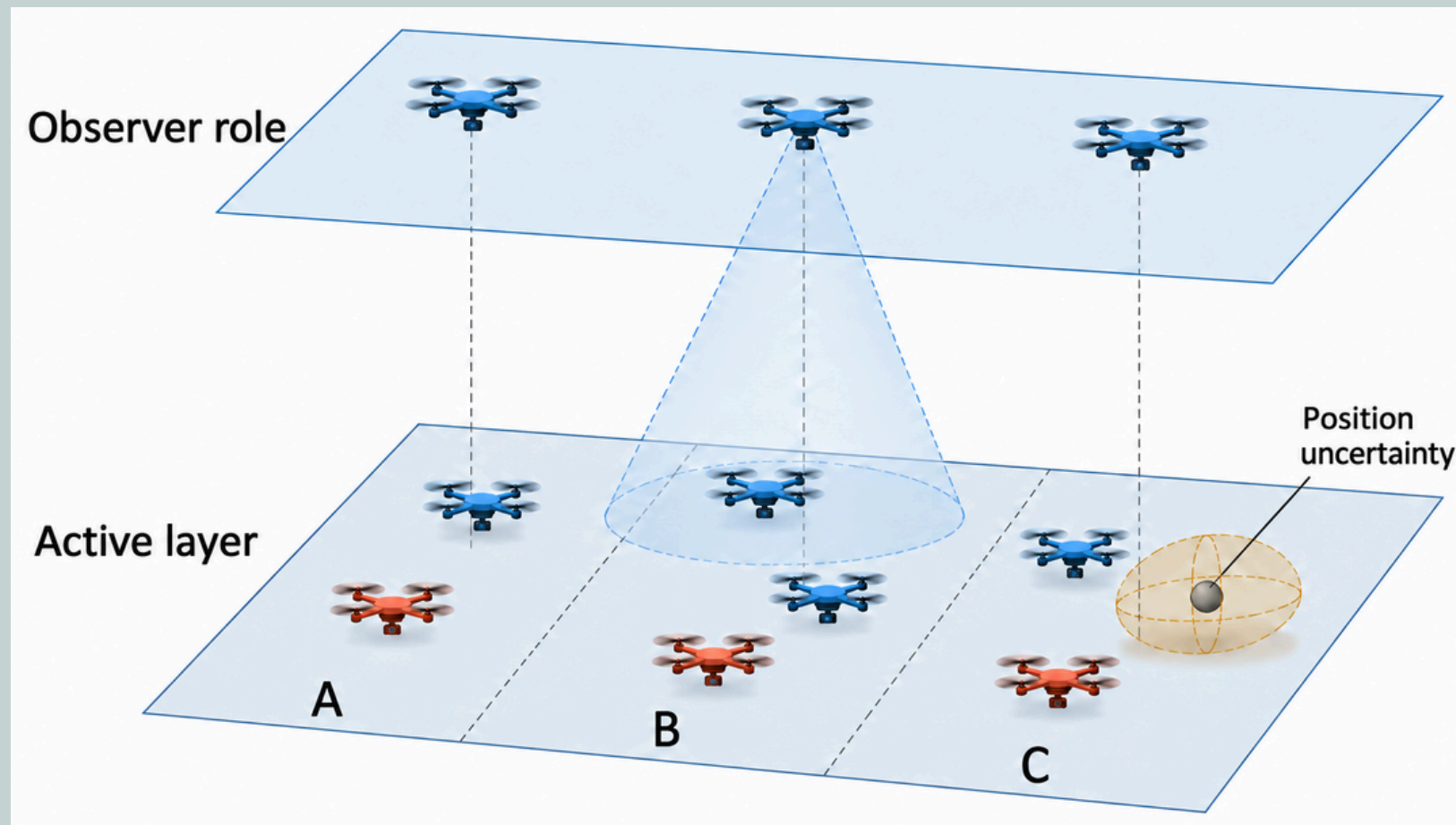
>> *RSAF Air Defence and Operations Command*

>> *DSTA and DSO*

With One Main Goal

To defend a densely populated city

Drones - Interceptor and Observer



Interceptor

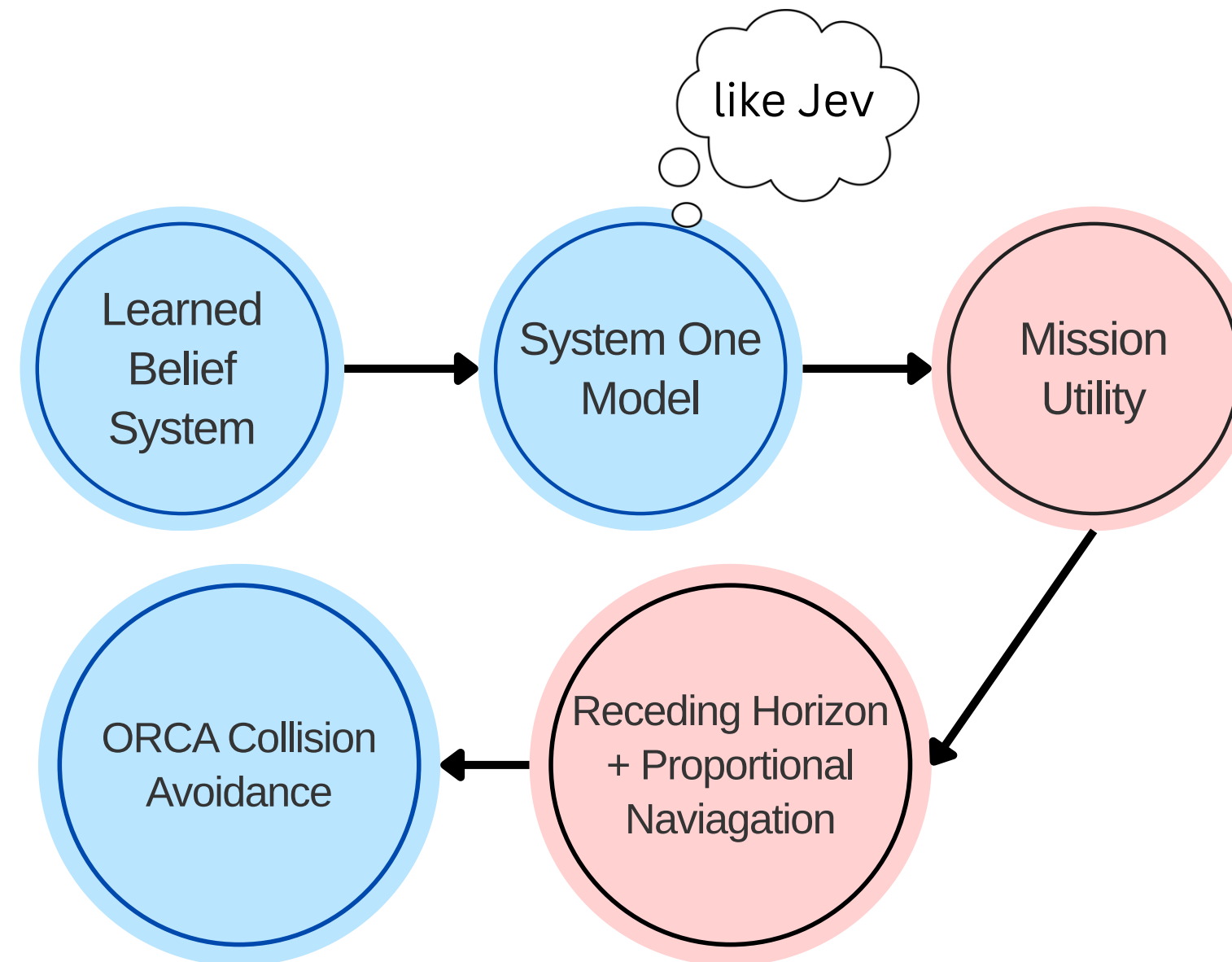
Carries out the interception using its own local observations.

Observer

Watches the interceptor and target from above, maintains awareness, and can become a backup interceptor after a failed attempt.

Each drone decides locally! No central dispatcher or radio handoff.

How it works



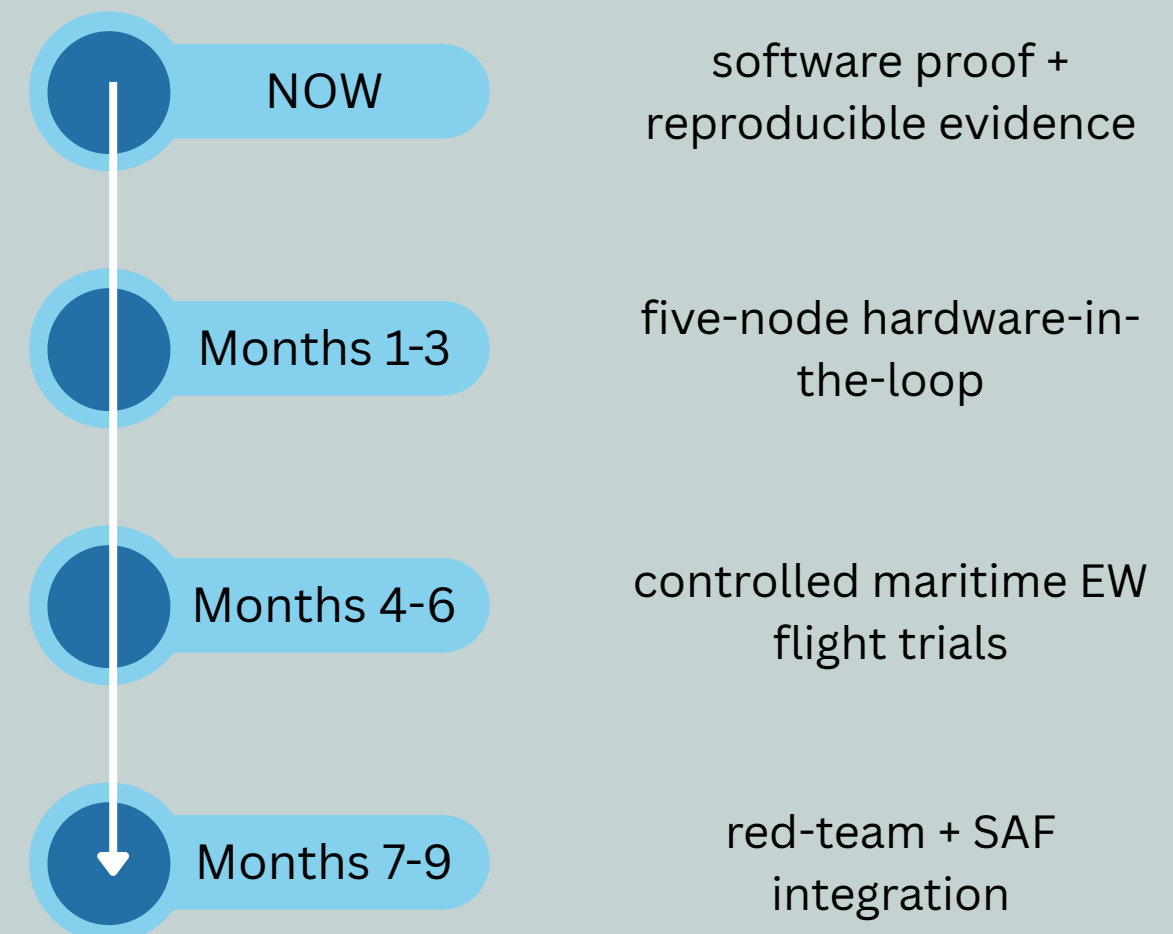
Limits

1. Harsh Weather
2. Night Operation
3. Non-Zero Leakage

Costs

~ \$1,000 modeled airframe + compute
+ dual optical seekers

Our RoadMap



Thank you